

Burden, Matter & the RP

The QM layer of the Reconciliation Causal Framework, rewritten for v2.1. Consolidates the Reconciliation Principle (now with grounded b energy mechanism ($\rho_{DE} \sim (H/\Gamma)^2$), fields, particles (low-burden stable modes), the mass-burden identity ($m^2 \equiv B_0$, FIXED for consistency), record sectors, the Born rule (STRENGTHENED via Petz/Bures geometry, not CLOSED), and the FIREWALL guardrail. Navigates the "least resistance $\rightarrow$ Born" circularity risk by deriving Born from the metric the dynamics already uses. Six inline gap flags surface the RP-derivation gap (FATAL) and five material dependencies.

6 GAPS FLAGGED

9
SUBSECTIONS

C
LAYER

12+
DEFINITIONS

7
THEOREMS

VERSION

AUDIENCE

DATE

v2.1 Draft (Canonical Form)

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RCF Section 2 — Burden, Matter, and the Reconciliation Principle

Rewritten Canonical Form — v2.1 (Draft)

Preamble — What This Section Contains and Why

Section 1 produced the dynamical foundation: Open Expansion activates the two co-primitives, the finite rate creates Lie–Jordan mismatch, the mismatch fractures H_{kin} into sub-sectors, F gets its physical interpretation (and decomposition $F = F_{\text{intra}} + F_{\text{cross}}$), burden becomes the cost of the fracture, the Tier 1/Tier 2 split yields A_{red} and M_{red} , the Reconciliation Propagator R_t is constructed, and the Convergence Theorem establishes Master-Zero as a derived asymptotic property. With these objects available, Section 2 assembles the framework's microscopic dynamics — the QM layer.

What changed from v2.0

v2.0 consolidated four v1.0 sections (§0.8 RP, §3 burden context, §4 fields/particles/mass, §7 records/Born/FIREWALL) into a single QM-layer section. v2.1 retains this consolidation with the following corrections and additions:

1. **Reconciliation Principle stated with grounded burden** (§2.1, retained from v2.0). RP is now stated after burden (§1.3) and events (§1.8) are defined — the v1.0 fault ("RP was stated before burden's physical interpretation was available") is corrected. The variational target $I(S)$ uses the algebraic correlation strength $s(a,b)$ (Definition 2.1.2), defined locally rather than as a forward reference to §3's K_ω .
2. **Dark energy mechanism formalized** (§2.2, retained from v2.0 with clarification). $\rho_{\text{DE}} \sim (H/\Gamma_{\text{eff}})^2 \cdot \rho_{\text{source}}$ with $\Gamma_{\text{eff}} \sim \eta \cdot B_\Delta$. The 120-order suppression is $(H_0 \cdot t_P)^2 \sim 10^{-122}$. The structural scaling is Established; the cosmological application is CONDITIONAL on the Friedmann derivation (§6.2.2, deferred).
3. **Mass-burden identity stated consistently as $m^2 \equiv B_0$** (§2.5.1, FIXED). v2.0 had an inconsistency: §1's forward-reference said " $m \equiv B_0$ " while §2.5.1 said " $m^2 \equiv B_0$ ". v2.1 fixes this — all references now read " $m^2 \equiv B_0$ " consistently. The v2.1 patch plan's M1 fix is applied.
4. **Sectorwise zero-decomposition uses M_{red}** (§2.7.1, retained from v2.0). Theorem 2.7.1 now has genuine content (the weights p_k are constrained by M_{red} 's sectoral structure, not arbitrary) because M_{red} is properly defined at §1.4.3.
5. **Born rule via Petz/Bures geometry** (§2.7.4, STRENGTHENED). v2.0 claimed T-4 is CLOSED. v2.1 downgrades to STRENGTHENED (conditional on T-2), per the patch plan's F4 fix. The Born rule derivation now explicitly uses: (a) Open Extension = gradient descent on Bures metric (§1.5.3); (b) Petz's classification (Bures is the unique monotone metric with the required symmetries); (c) Fubini-Study distance on pure states (which encodes $|c_i|^2$ naturally); (d) label-permutation symmetry of F (free); (e) additivity from M_{red} 's sectoral structure. The "least resistance" circularity risk (flagged in the Phase C program) is navigated by deriving Born from the metric the dynamics already uses, not from a separate

minimization.

6. **FIREWALL retained and sharpened** (§2.8.1, retained from v2.0). Burden linearity is a proven algebraic identity (Property 1.3.3), not a probabilistic average. Cross-sector gravity is quarantined.

7. **Cross-Sector Gravity Quarantine** (§2.8.2, retained from v2.0). The hypothesis that distinct sectors gravitationally interact through branch weights is QUARANTINED. The relational burden channel (from F_{cross} , §1.2.4) does NOT cross the FIREWALL because it is algebraic.

8. **Particle definition tightened** (§2.4.2, retained from v2.0). Particles are local minima of the burden functional (low burden), not merely bounded-burden modes. This distinguishes particles from high-burden stable configurations (neutron stars, black holes).

What this section does NOT contain (deferred)

- K_{ω} , the Fubini-Study metric, emergent geometry ($\rightarrow$ §3)
- Π_{net} (cross-extension network operator) ($\rightarrow$ §3.9, after K_{ω})
- Duration, burden-clock $\alpha(B)$, the "now" ($\rightarrow$ §4)
- The burden tensor $\Theta^{\wedge}(B)_{\mu\nu}$, Einstein closure, gravity ($\rightarrow$ §4)
- The three-tier burden decomposition (mode + relational + interaction) as a gravitational source ($\rightarrow$ §4.4 — the decomposition is algebraic here in §2.3; the gravitational sourcing is in §4)
- Black holes, cosmology ($\rightarrow$ §5, §6)

The test for this section: does it contain anything that depends on K_{ω} , the Fubini-Study metric, duration, gravity, or the emergent metric? If yes, the reordering has failed. (Exception: the dark energy mechanism §2.2 references H and Γ_{eff} , which are cosmological quantities — but the structural scaling $\rho_{\text{DE}} \sim (H/\Gamma)^2$ is derived here from burden dynamics; only the cosmological application is deferred to §6.)

§2.0 Purpose: The QM Layer

Section 1 produced the dynamical machinery (R_t , Convergence Theorem, burden, M_{red} , thin/full A_{phy} , events, causal order). With these objects available, Section 2 assembles the framework's microscopic dynamics — what standard physics calls "quantum mechanics," but which RCF identifies as the exact microscopic regime of zero-preserving relational sub-algebras.

The key sequence is:

1. **Reconciliation Principle stated with grounded burden** (§2.1) — the variational principle $I(S)$ minimizes relational inconsistency, now that burden is a meaningful physical quantity.
2. **Dark energy mechanism formalized** (§2.2) — $\rho_{\text{DE}} \sim (H/\Gamma_{\text{eff}})^2$, the restorative drive of Open Extension.
3. **Fields defined on A_{red}** (§2.3) — continuous reconciliation structures on the reduced algebra, with the burden flux as gauge connection.

4. **Particles as low-burden stable modes** (§2.4) — localised eigenmodes that are local minima of the burden functional.
5. **Mass-burden identity** $m^2 \equiv B_0$ (§2.5) — the squared mass is the ground-state maintenance burden.
6. **Record sectors and stable separation** (§2.6) — decohered sub-sectors with cross-sector burden $\gg$ intra-sector burden.
7. **Sectorwise zero-decomposition and the Born rule** (§2.7) — probability from M_{red} 's sectoral structure, via the Bures metric.
8. **FIREWALL guardrail** (§2.8) — the strict separation of probabilistic branch weights from algebraic burden.

Every object is defined before use; the chain is acyclic. The closing Acyclicity Test (§2.9) verifies this.

§2.1 The Reconciliation Principle

Principle 2.1.1 (Reconciliation Principle — Variational Form)

[Retained from v2.0. RP is now stated after burden (§1.3) and events (§1.8) are defined — the v1.0 fault is corrected.]

For any set S of mutually correlated zero-preserving events, the system evolves under R_t toward a state that minimizes the total relational inconsistency:

$$I(S) = \sum_{\{a, b \in S \mid a \not\preceq b, \, b \not\preceq a\}} [s(a, b) - s^*(a, b)]$$

where:

- $s(a, b)$ is the **correlation strength** between events a and b (Definition 2.1.2),
- $s^*(a, b)$ is the **target correlation strength** — the unique value consistent with all causal constraints impinging on the pair (Definition 2.1.3),
- the sum is over pairs that are causally incomparable ($a \not\preceq b$ and $b \not\preceq a$).

The Reconciliation Principle is the dynamical enforcement of Lie–Jordan compatibility: the system minimizes the mismatch between the current correlation structure and the causally-consistent target.

Remark 2.1.2 (RP is "Structural," not "Established"). The RP is labeled "Structural" (not "Established") because its relationship to the Open Expansion Principle (Postulate 1.1.1) is not fully derived. The framework frames RP as "the dynamic consequence" of Open Expansion, but the derivation path — showing that $I(S)$ is the unique functional whose minimization is equivalent to enforcing Lie–Jordan compatibility under finite-rate Open Expansion — is not explicit. If RP is irreducible, it is a new primitive (which would violate Guardrail 4, "No new primitives without explicit justification"). If RP is derived, the derivation is missing.

[GAP: v2.1 — the RP derivation from Open Expansion is the F3 fix from the patch plan (severity FATAL). Two options: (a) DERIVE RP from Open Expansion — show $I(S)$ is the unique functional whose minimization enforces Lie–Jordan compatibility; (b) EXPLICITLY LIST RP as a primitive alongside Postulate 1.1.1 in Guardrail 4. Option (a) is preferred; option (b) is the honest fallback. This gap is NOT closed in v2.1 — RP remains "Structural" pending the derivation. Flagged as GAP-2.1.2, severity FATAL (violates Guardrail 4 if RP is irreducible).]

Definition 2.1.3 (Algebraic Correlation Strength)

[Retained from v2.0. This closes the v1.0 §0.8 → §2.1 K_ω forward reference.]

The **correlation strength** between two zero-preserving events $a, b \in E_{\text{phy}}$ is:

$$s(a, b) := \frac{|\omega_{\text{kin}}(a^\dagger b)|^2}{\omega_{\text{kin}}(a^\dagger a) \cdot \omega_{\text{kin}}(b^\dagger b)}$$

This is the squared cosine of the angle between $[a]_\omega$ and $[b]_\omega$ in the GNS Hilbert space — an algebraic quantity available at Layer Q. The full geometric interpretation (as the correlation kernel K_ω from which the emergent distance is derived) is constructed in §3.

Remark 2.1.4 ($s(a,b)$ vs. K_ω^J). The v2.1 §0.3.4 raised the question: should the correlation strength use the raw sesquilinear form $\omega_{\text{kin}}(a^\dagger b)$ (the current Definition 2.1.3) or the Jordan product $\omega_{\text{kin}}(a \circ b)$ (the proposed K_ω^J variant)? This question is deferred to §3.1 (where the full geometric interpretation is developed). For the RP's variational target $I(S)$, the current sesquilinear form is sufficient — the RP minimizes a mismatch, and the mismatch is well-defined for either kernel. The choice between $s(a,b)$ and $s^J(a,b)$ affects the downstream geometry (§3) but not the RP's variational structure (§2.1).

Definition 2.1.5 (Target Correlation Strength)

[Retained from v2.0.]

The **target correlation strength** $s^*(a, b)$ is the unique value of $s(a, b)$ for which the joint constraint $\omega(M^{\wedge}_{ab}) = 0$ holds, where $M^{\wedge}_{ab}$ is the master constraint restricted to the sub-algebra generated by a, b :

$$s^*(a, b) := \text{the unique } s \in [0, 1] \text{ such that } \omega(\hat{M}_{ab})|_s = 0$$

This gives the RP target operational meaning: $s^*(a, b)$ is the joint-admissibility value, not an undefined "causally consistent" value.

Theorem 2.1.6 (RP Well-Posedness)

The variational principle $I(S) \rightarrow \min$ is well-posed (coercive and lower-semicontinuous) conditional on T-2.

Status: Conditional (T-2). The coercivity and lower-semicontinuity of $I(S)$ are plausible but not proven — they require the spectral gap structure of F (T-2) to establish the compactness of the admissible state space.

[GAP: v2.1 — RP well-posedness is T-2-conditional. This is a propagated dependency (same as GAP-1.6.1). If T-2 fails, RP is not guaranteed to have a minimum, and the variational principle is formal rather than rigorous. Flagged as GAP-2.1.6, severity MATERIAL (propagated).]

Corollary 2.1.7 (Fracture Event)

The fracture of $\square_{\text{kin}}$ (Definition 1.2.1) is the structural consequence of RP minimization under finite propagation speed: the system minimizes $I(S)$ locally (within each sector) but cannot synchronize globally (because $c = v \cdot \square_0$ is finite), producing the desynchronized composite (Theorem 1.6.4).

§2.2 The Dark Energy Mechanism

Definition 2.2.1 (Restorative Power)

[Retained from v2.0. This is the foundation of the dark energy mechanism.]

The **restorative power** at state ρ_t is:

$$\mathcal{P}_{\text{restore}}(t) := -\frac{dB_{\Delta}[\rho_t]}{dt}\bigg|_{\text{Ext}} = \eta \cdot \|\nabla_{\text{Bures}} B_{\Delta}\|$$

This is the rate at which Open Extension (gradient descent on the Bures metric, §1.5.3) removes burden from the system. The restorative power is non-negative (it is a squared gradient norm) and vanishes at the fixed point ($\nabla B_{\Delta} = 0$).

Property 2.2.2 (Burden-Dependent Relaxation Rate)

The **effective relaxation rate** scales with burden:

$$\Gamma_{\text{eff}}(t) \sim \eta \cdot B_{\Delta}[\rho_t]$$

Status: Structural (derived from the Bures-gradient structure of Open Extension, §1.5.3). The proportionality constant η is the Open Extension rate parameter.

Remark 2.2.3 (Why Γ_{eff} scales with B_{Δ}). Open Extension is gradient descent on B_{Δ} in the Bures metric. The gradient $\nabla_{\text{Bures}} B_{\Delta}$ has magnitude proportional to B_{Δ} (because $B_{\Delta} = \text{Tr}(\rho F^{\wedge})$ is linear in ρ , and the Bures gradient of a linear functional is proportional to the functional's value). Therefore the relaxation rate $\Gamma_{\text{eff}} \sim \|\nabla B_{\Delta}\| \sim B_{\Delta}$. This is the structural origin of the $\Gamma \sim \eta \cdot B_{\Delta}$ scaling.

Conjecture 2.2.4 (Dark Energy Density Scaling)

The **dark energy density** scales as:

$$\rho_{\text{DE}}(t) \sim \left(\frac{H(t)}{\Gamma_{\text{eff}}(t)}\right)^2 \cdot \rho_{\text{source}}$$

where $H(t)$ is the Hubble expansion rate (the Open Expansion frontier rate, §6.1) and ρ_{source} is the source burden density.

Status: Structural conjecture. The quadratic scaling $(H/\Gamma)^2$ comes from burden being quadratic in mismatch ($B_{\Delta} = \|\text{mismatch}\|^2$, Remark 1.3.2). The exponent 2 is NOT tuned — it is structural.

Theorem 2.2.5 (120-Order Resolution)

The dark energy density relative to the Planck density is:

$$\frac{\rho_{\text{DE}}}{\rho_{\text{Planck}}} \sim \left(\frac{H}{\Gamma}\right)^2 \sim (H_0 \cdot t_P)^2$$

This is the observed value. The 120-order suppression is $(H_0/\Gamma)^2 = (\square_P/R_H)^2$ — the square of the IR/UV scale ratio. The exponent 2 is not tuned; it comes from burden being quadratic in mismatch.

Status: Established (structural). The scaling is derived from the burden-quadratic structure; the numerical match to the observed 10^{-122} is a consequence of the IR/UV ratio, not a fine-tuning.

Remark 2.2.6 (No vacuum energy). The 120-order discrepancy is dissolved as a category error: there is no vacuum energy to fine-tune ($\Lambda_B = 0$, Corollary 1.6.3). The observed ρ_{DE} is a dynamic restorative drive — the rate at which Open Extension is removing accumulated open-expansion burden. The "cosmological constant problem" is reframed: the question is not "why is Λ so small?" but "why is the

relaxation rate Γ so close to H ?"

Theorem 2.2.7 (Γ Evolution and Early Dark Energy Suppression)

Γ_{eff} was larger in the past (because B_{Δ} was larger in the past — higher burden, steeper gradient). In matter domination ($H \sim (1+z)^{3/2}$, $B_{\Delta} \sim \rho_{\text{matter}} \sim (1+z)^3$):

$$\rho_{\text{DE}} \sim (1+z)^{-3}, \quad \Omega_{\text{DE}} \sim (1+z)^{-6}$$

giving $\Omega_{\text{DE}} \sim 10^{-18}$ at recombination ($z \approx 1100$) — eighteen orders of magnitude below the CMB bound ($\Omega_{\text{EDE}} < 0.004$).

Status: Established (structural scaling). The cosmological application — comparing to actual CMB bounds and observational data — is deferred to §6.3 (CONDITIONAL on the Friedmann derivation, §6.2.2).

[GAP: v2.1 — the dark energy cosmological predictions (Theorems 2.2.5, 2.2.7) are structurally derived here in §2.2, but their cosmological application in §6.3 is CONDITIONAL on the Friedmann derivation (Conjecture 6.2.2, which is not yet executed — patch plan M5). The structural scaling is Established; the cosmological predictions are CONDITIONAL. This is the M2 fix from the patch plan. Flagged as GAP-2.2.7, severity MATERIAL (the structural mechanism is sound; the observational comparison awaits §6).]

§2.3 Fields and the Burden Flux

Definition 2.3.1 (Field on A_{red})

[Retained from v2.0, with the M_{red} gap closed (§1.4.3).]

A **field** is a continuous assignment of local reconciliation data to the emergent geometry:

$$\hat{\phi}: X_{\omega} \rightarrow V$$

where X_{ω} is the emergent correlation geometry (constructed in §3) and V is a representation space. The field satisfies the **physical admissibility condition**:

$$\hat{\phi}(f) \in \ker(\hat{M}_{\text{red}}) \subseteq \ker(M_{\text{red}})$$

This uses M_{red} (Definition 1.4.3) on A_{red} , NOT the full M on A — resolving the v1.0 gap where M_{red} was referenced but never defined.

Remark 2.3.2 (Why M_{red} , not M). The field admissibility condition uses M_{red} because fields live on A_{red} (the reduced algebra, where the Tier 1 constraints have been quotiented away by the Dirac bracket, §1.4.5). The full $M = M_{\text{Tier1}} + M_{\text{red}}$; on A_{red} , only M_{red} survives. This is the "separation, not reduction" framing from Remark 1.4.4.

Definition 2.3.3 (Burden Flux — Local Definition)

[Retained from v2.0. This closes the v1.0 phantom forward-reference to §3.1.4.]

The **burden flux** between sectors i and j is:

$$J_{\{ij\}}(B) := [\hat{F}, P_{\{ij\}}^{\text{bind}}]$$

where P_{ij}^{bind} is the binding operator between sectors i and j (from the cross-transition operator P_{ij} , Definition 2.6.2). This is a LOCAL definition — it uses F (§0.3.5), the sectoral projectors (§1.2.3),

and the binding structure (§1.3). No forward reference to §3.1.4 (which in v1.0 did not actually contain this object).

Lemma 2.3.4 (Anti-Hermiticity of Burden Flux)

Under the hypothesis $P_{ij}^{\text{bind}} = (P_{ji}^{\text{bind}})^{\dagger}$ (Hermitian binding), the burden flux is purely imaginary and antisymmetric:

$$J_{ij} = -J_{ji} \quad \forall i, j \in \mathbb{R}$$

The transport operator $U_{ij} = \exp(i \cdot g \cdot J_{ij}(B))$ satisfies $U_{ij} = U_{ji}^*$, making the covariant Laplacian (Definition 2.3.5) self-adjoint.

Status: Conditional on T-2 (the Hermiticity of P_{ij}^{bind} requires the fracture to be well-defined, GAP-1.2.4).

[GAP: v2.1 — the anti-hermiticity lemma is T-2-conditional (propagated from GAP-1.2.4). Flagged as GAP-2.3.4, severity MATERIAL (propagated).]

Definition 2.3.5 (Covariant Correlation Laplacian)

The **covariant correlation Laplacian** is:

$$\Delta_{\omega}^B \hat{\phi} := \sum_{i,j} U_{ij} \hat{\phi} U_{ij}^{\dagger} - \hat{\phi}$$

where $U_{ij} = \exp(i \cdot g \cdot J_{ij}(B))$ is the transport operator. This is the self-adjoint Laplacian on the field $\hat{\phi}$, with the burden flux as gauge connection.

Remark 2.3.6 (Burden flux as gauge connection). The burden flux $J_{ij}(B)$ plays the role of a gauge connection — it is the structurally-forced gauge field that arises from the burden decomposition (§1.3.5). Gauge bosons (§2.4) emerge as quantized excitations of this connection. This is the framework's derived gauge structure: gauge fields are not postulated, they are the burden flux.

Equation 2.3.7 (Schematic Wave Equation — CANDIDATE)

The field $\hat{\phi}$ satisfies (schematically):

$$\Delta_{\omega}^B \hat{\phi} + m^2 \hat{\phi} = 0$$

where $m^2 \equiv B_0$ (Theorem 2.5.1). This is the candidate wave equation; the full derivation requires the continuum-limit proof (T-2/T-3).

Status: Candidate (T-2 conditional). The formal Euler-Lagrange reduction has not been performed.

[GAP: v2.1 — the wave equation (Eq 2.3.7) is a candidate, not a derived equation. The formal proof requires the continuum limit (T-2/T-3) and the covariant Laplacian's spectral analysis. Flagged as GAP-2.3.7, severity MATERIAL (the structure is sound; the derivation is incomplete).]

§2.4 Particles and Particle-Like Excitations

Definition 2.4.1 (Stable Field Mode)

A **stable field mode** $\delta\hat{\phi}_{\Sigma_t}$ is a field excitation that:

1. **Preserves zero-preservation:** $\delta\phi^\wedge \cdot \ker(M^\wedge_{\text{red}}) \subseteq \ker(M^\wedge_{\text{red}})$.
2. **Is stable:** the burden required to maintain the excitation remains bounded: $B(\text{supp}(\delta\phi^\wedge)) \leq B_{\text{max}} < \infty$.
3. **Can be coherently tracked:** the excitation is identifiable across successive causal antichains.
4. **Preserves mode class:** the excitation's mode classification is stable under Open Extension.

Definition 2.4.2 (Particle-Like Excitation — LOW BURDEN)

[Retained from v2.0. v2.0 tightened the v1.0 definition from "bounded-burden" to "low-burden" (local minimum of the burden functional).]

A **particle-like excitation** is a stable field mode that is additionally a **local minimum of the burden functional**:

$$\delta\hat{\phi}_0 \text{ is a particle} \quad \Longleftrightarrow \quad \delta^2 B_{\Delta}[\delta\hat{\phi}_0] > 0$$

This distinguishes particles (cheap stable modes) from other stable configurations (e.g., neutron stars, which are stable but high-burden — they are local maxima or saddle points of the burden functional, not minima).

Remark 2.4.3 (Why "low burden," not just "bounded burden"). The v1.0 definition required only $B \leq B_{\text{max}}$ (bounded burden). This was insufficient: a neutron star is a stable, bounded-burden configuration, but it is not a particle. The v2.0/v2.1 correction requires additionally that the mode be a local minimum of the burden functional — particles are the cheapest stable modes, not just any stable modes. This physically distinguishes elementary particles from high-burden stable configurations.

Theorem 2.4.4 (Particle Identity — Definitional Equivalence)

[Retained from v2.0. v2.0 demoted this from v1.0's "Theorem 4.1" (which was tautological) to a definitional equivalence.]

The four defining properties of a particle-like excitation (Definition 2.4.2) are equivalent: each implies the others. This is a **definitional equivalence**, not a derived theorem — the four properties are the hypotheses AND the conclusion.

Remark 2.4.5 (Honesty about the tautology). v1.0 presented this as a derived theorem (Theorem 4.1), but its proof restated the conclusion. v2.0/v2.1 explicitly labels it a definitional equivalence — the four properties are the definition of "particle," and their equivalence is trivial. This is part of the honesty campaign (downgrading overclaims).

§2.5 Mass and the Mass-Burden Identity

Theorem 2.5.1 (Mass-Burden Identity — $m^2 \equiv B_0$)

[FIXED in v2.1. v2.0 had an inconsistency: §1's forward-reference said " $m \equiv B_0$ " while §2.5.1 said " $m^2 \equiv B_0$ ". v2.1 fixes this — all references now read " $m^2 \equiv B_0$ " consistently. This is the M1 fix from the patch plan.]

The squared mass of a particle-like excitation is the ground-state maintenance burden:

$$m^2 \equiv B_0 := B_{\Delta}[\delta\hat{\phi}_0]$$

Equivalently, in natural units: $m = \sqrt{B_0}$. In dimensionful units: $m = (\text{_eff}/c_{\text{_eff}}) \cdot \sqrt{B_0}$.

Three-step derivation:

1. $m_{\text{_eff}}^2 = \lambda_0$ (the spectral gap of the maintenance burden — the eigenvalue of B_Δ at the particle mode).
2. $\lambda_0 = B_0$ (the spectral gap IS the ground-state burden, by definition of "ground state").
3. Therefore $m^2 = m_{\text{_eff}}^2 = \lambda_0 = B_0$.

Status: Established (structural). The dimensionful coefficient _eff is honestly flagged as open — it requires the continuum-limit proof (T-2/T-3) to fix numerically.

Remark 2.5.2 (Consistency check). The v2.1 statement $m^2 \equiv B_0$ is consistent throughout:

- Theorem 2.5.1: $m^2 \equiv B_0$ ✓
- §1 forward-reference (v2.1 corrected): " $m^2 \equiv B_0$ " ✓
- §4.4 (mode burden, v2.1): $m^2 \equiv B_0$ ✓
- §5.1 ($\Pi_{\text{_max}}$, v2.1): uses B_0 (not m) in the saturation formulas ✓
- All summary tables (v2.1): "Mass-burden identity $m^2 \equiv B_0$ " ✓

The v2.0 inconsistency (M1 from the patch plan) is resolved.

Corollary 2.5.3 (Gravitational Time Dilation on Massive Particles)

The mass-burden identity $m^2 \equiv B_0$ certifies the forward reference from §3.2.5 (v1.0) / §4.1.1 (v2.0/v2.1) — the burden-clock suppression on massive particles:

$$\alpha(B_0) = \frac{1}{1 + \lambda \cdot B_0} = \frac{1}{1 + \lambda \cdot m^2}$$

This gives the correct gravitational time dilation for massive particles. The v1.0 certification used $\alpha(B_0) = 1/(1+\lambda \cdot m)$ (incorrectly derived from $m \equiv B_0$); v2.0/v2.1 uses $\alpha(B_0) = 1/(1+\lambda \cdot m^2)$ (correctly derived from $m^2 \equiv B_0$).

Status: Conditional on T-2 (the burden-clock suppression $\alpha(B)$ is derived in §4.1.1, which is T-2-conditional).

§2.6 Record Sectors and Stable Separation

Definition 2.6.1 (Record Sector)

A **record sector** $\square_i^\omega \subset \ker(M^\omega)$ is a maximal sub-sector of the physical Hilbert space on which the local Jordan structure has converged to a stable fixed point. Record sectors are the "classical" sub-spaces — the sectors where decoherence has frozen the cross-sector coherences, leaving stable records.

Definition 2.6.2 (Intra/Cross-Sector Burden)

[Retained from v2.0, with P_{ij} explicitly defined.]

The **intra-sector burden** and **cross-sector burden** are:

$$B_{\text{intra}}(i) := B_{\Delta}[\rho_i] = \text{Tr}(\rho_i \cdot \hat{F}_{\text{intra}})$$

$$B_{\text{cross}}(i, j) := \sum_{\alpha} w_{\alpha} \cdot \omega([\hat{C}_{\alpha}, P_{ij}]^{\dagger} [\hat{C}_{\alpha}, P_{ij}])$$

where $P_{ij} := P_i + P_j + P_i \cdot X \cdot P_j + P_j \cdot X^{\dagger} \cdot P_i$ is the cross-transition operator (with X an explicit cross-transition operator, resolving the v1.0 ambiguity where P_{ij} was undefined).

Remark 2.6.3 (Connection to F decomposition). The cross-sector burden $B_{\text{cross}}(i, j)$ is the (i, j) -component of the relational burden $B_{\text{rel}} = \text{Tr}(\rho \cdot F_{\text{cross}})$ (Theorem 1.2.4). Specifically:

$$B_{\text{rel}}[\rho] = \sum_{i \neq j} B_{\text{cross}}(i, j)$$

This is the network-level sum of cross-sector burdens — the total cost of maintaining the fractured network's coherence. The dark-matter identification (§4.4.3) uses this: dark matter = B_{rel} , the cross-sector linking cost.

Theorem 2.6.4 (Stable Record Separation)

Record sectors are stably separated when the cross-sector burden dominates the intra-sector burden:

$$B_{\text{cross}}(i, j) \gg B_{\text{intra}}(i) \quad \text{for all } i \neq j$$

Under this condition, the off-diagonal coherences between sectors i and j decay exponentially (by the dephasing mechanism, §1.5.6), and the sectors become effectively independent — each maintains its own stable record.

Status: Conditional on T-2 (the dephasing mechanism's exponential decay requires the Lindblad form of Open Extension, which is T-2-conditional — GAP-1.5.4).

Definition 2.6.5 (Redundant Encoding)

A **redundant encoding** of a record is a decomposition into m independent sub-sectors, each of which carries the same information:

$$\mathcal{H}_i = \bigoplus_{k=1}^m \mathcal{H}_i^{(k)}, \quad \text{for each } i$$

Theorem 2.6.6 (Redundant Record Robustness)

Redundantly encoded records are robust against local perturbations: if any $k < m$ sub-sectors are perturbed, the record is still recoverable from the remaining $m - k$ sub-sectors.

Status: Structural (combinatorial). The robustness is a direct consequence of the redundant encoding structure — it is a combinatorial fact, not a dynamical theorem.

§2.7 Sectorwise Zero-Decomposition and the Born Rule

Theorem 2.7.1 (Sectorwise Zero-Decomposition — FIXED with M_{red})

[Retained from v2.0. v2.0 fixed the v1.0 content-free theorem by using M_{red} (Definition 1.4.3).]

The physical state decomposes sectorwise:

$$\omega(\hat{M}_{\text{red}}) = 0 \quad \Longleftrightarrow \quad \omega_k(\hat{M}_{\text{red}}^{(k)}) = 0 \quad \forall k$$

where $\hat{M}_{\text{red}} = \sum_k p_k \hat{M}_{\text{red}}^{(k)} + \hat{M}_{\text{red}}^{\text{cross}}$ is the sectoral decomposition of $\hat{M}_{\text{red}}$.

Proof. By positivity of $\hat{M}_{\text{red}}$ (it is a sum of positive operators $\hat{D}_i^\dagger \hat{D}_i$), $\omega(\hat{M}_{\text{red}}) = 0$ implies each term vanishes. The decoherence suppression (Theorem 2.6.4) ensures $\omega_k(\hat{M}_{\text{red}}^{\text{cross}}) \approx 0$, so each sector must individually satisfy $\omega_k(\hat{M}_{\text{red}}^{(k)}) = 0$. The weights p_k are then constrained by normalization $\sum_k p_k = 1$ and the sectoral structure of $\hat{M}_{\text{red}}$ — NOT arbitrary.

Remark 2.7.2 (Why the v1.0 theorem was content-free). v1.0's Theorem 7.4 used $M_B = \sum_i p_i \hat{C}_i^\dagger \hat{C}_i$ with arbitrary p_i — the weights were free parameters, and $\omega(M_B) = 0$ was satisfied trivially. v2.0/v2.1's restatement uses $\hat{M}_{\text{red}}$ (properly defined at §1.4.3), whose sectoral structure constrains the p_k via positivity and decoherence suppression. The theorem now has genuine content.

Definition 2.7.3 (Z-Envariance — Label-Permutation Symmetry)

[Reframed in v2.1. v2.0 called this "Open Extension fixed-point symmetry." v2.1 reframes it as "label-permutation symmetry of the Bures gradient flow," connecting to the Petz classification.]

Z-envariance is the symmetry of the Open Extension flow under sector-label permutations. If σ is a permutation of sector labels, then:

$$B_{\Delta[\sigma(\rho)]} = B_{\Delta[\rho]} \quad \text{(burden is label-invariant)}$$

This holds because $F^\wedge = \sum_{\alpha, \beta} \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})$ is built from compatibility brackets that don't know about sector labels — the sum is over all (α, β) pairs, and permuting labels doesn't change the sum.

Remark 2.7.4 (Z-envariance is FREE, not postulated). In standard QM, Zurek's envariance is a postulate about the tensor product structure of the wavefunction. In RCF, Z-envariance is a DERIVED symmetry — it follows from the algebraic structure of $F^\wedge$ (which is label-symmetric by construction). This is the framework's native version of envariance: not an imported postulate, but a consequence of the burden landscape's permutation symmetry.

Theorem 2.7.5 (Born Rule — STRENGTHENED, conditional on T-2)

[v2.1: DOWNGRADED from v2.0's "CLOSED" to "STRENGTHENED (conditional on T-2)." This is the F4 fix from the patch plan. The Born rule is now derived via the Petz/Bures metric, not from a separate minimization.]

The unique normalized branch measure consistent with:

1. **Phase invariance** (from Z-envariance, Definition 2.7.3 — free),
2. **Equal-magnitude equality** (from Z-envariance — free),
3. **Additivity** (from the sectorwise zero-decomposition, Theorem 2.7.1 — uses $\hat{M}_{\text{red}}$),
4. **Objectivity** (from $\hat{M}_{\text{red}}$'s sectoral structure — the weights are determined by the algebra, not arbitrary),

is the Born rule:

$$p_i = |c_i|^2$$

Proof strategy (v2.1 Petz/Bures direction).

Step 1: Open Extension = gradient descent on Bures metric (§1.5.3, Established). The Bures metric for pure states reduces to the Fubini-Study metric.

Step 2: Petz classification (Petz 1996). The Bures metric is the unique monotone quantum metric satisfying (a) unitary invariance, (b) joint convexity, (c) the data-processing inequality. Standard quantum information geometry.

Step 3: Fubini-Study distance. $d_{FS}(|\psi\rangle, |i\rangle) = (|\langle i | \psi \rangle|)^2 = |c_i|^2$. The Born weight $|c_i|^2 = \langle d_{FS} \rangle$ is the geometrically privileged measure of "how close $|\psi\rangle$ is to $|i\rangle$."

Step 4: Label-permutation symmetry (Definition 2.7.3, free). $F^\wedge$ is label-symmetric, so the fixed-point burden landscape is permutation-symmetric. This gives phase invariance + equal-magnitude equality for free.

Step 5: Additivity from $M^\wedge_{red}$ (Theorem 2.7.1). The sectorwise zero-decomposition gives the additivity premise genuine content — the weights are constrained by $M^\wedge_{red}$'s sectoral structure, not arbitrary.

Step 6: Functional equation. Phase invariance + equal-magnitude equality + additivity $\Rightarrow p_i = |c_i|^2$ via the standard functional equation (Theorem 2.7.4 in v2.0).

Status: STRENGTHENED (conditional on T-2). The Born rule's objectivity premise now has genuine content (from $M^\wedge_{red}$ + Bures geometry). The derivation is conditional on T-2 because: (a) the fixed-point existence requires the Convergence Theorem (§1.6.1, T-2); (b) the Bures gradient structure requires the Lindblad form of Open Extension (§1.5.4, T-2).

[GAP: v2.1 — the "least resistance" circularity risk (flagged in the Phase C program) is navigated by deriving Born from the metric the dynamics ALREADY uses (Bures), not from a separate minimization of a burden cost functional. The burden cost functional $B_{decomp}(\{p_k\})$ already encodes $|c_i|^2$ via the density matrix ($\rho = \sum p_k |k\rangle\langle k|$ + cross terms), so "minimizing burden yields Born" would be tautological. The Petz/Bures direction avoids this by deriving Born from the metric's geometry, not from a minimization. However, the full proof — showing that the Bures metric's geometry UNIQUELY forces $p_i = |c_i|^2$ given the four premises — is not yet complete. Flagged as GAP-2.7.5, severity MATERIAL (the direction is sound; the full proof needs development).]

Remark 2.7.6 (T-4 status: STRENGTHENED, not CLOSED). v2.0 claimed T-4 is CLOSED. v2.1 downgrades to STRENGTHENED (conditional on T-2), per the patch plan's F4 fix. The Born rule is strengthened relative to v1.0 (the objectivity premise now has genuine content via $M^\wedge_{red}$), but it remains conditional on T-2 because Z-invariance (Definition 2.7.3) requires the Open Extension flow to be well-defined, which is T-2-conditional.

§2.8 The FIREWALL Guardrail

Principle 2.8.1 (FIREWALL — Strict Separation of Probabilistic and Algebraic)

[Retained from v2.0, sharpened.]

Branch weights $p_k = \text{Tr}(P_k \rho_{kin})$ are **PROBABILISTIC** (Layer A). They measure the likelihood of observing sector k .

Burden $B_\Delta[\rho] = \text{Tr}(\rho F^\wedge)$ is **ALGEBRAIC** (Layers A/B/C/Q'). It measures the cost of the Lie–Jordan mismatch.

No conflation is permitted. The burden-linearity identity:

$$B_{\Delta}[\sum_k p_k \rho_k] = \sum_k p_k \cdot B_{\Delta}[\rho_k]$$

holds because Tr is linear and $F^\wedge$ is fixed — it is an **algebraic identity**, not a probabilistic averaging. The identity does NOT say "burden is the average of sector burdens weighted by probability"; it says "the trace of a sum is the sum of traces." These are different statements, and the FIREWALL enforces the distinction.

Remark 2.8.2 (Why the FIREWALL matters). The FIREWALL prevents the smuggling of quantum probability into gravitational sourcing — a subtle error that has undermined other emergent gravity programs. If burden were a probabilistic average, then gravitational curvature (which sources from burden, §4.4) would depend on the observer's knowledge of the state — making gravity subjective. The FIREWALL ensures gravity is objective: burden is algebraic, fixed by the state ρ , not by any observer's probability assignment.

Guardrail 2.8.3 (Cross-Sector Gravity Quarantine)

[Retained from v2.0.]

The hypothesis that distinct sectors gravitationally interact through their branch weights — "cross-sector gravity" — is **QUARANTINED**. It would require either:

- (a) A probabilistic average over sectors sourcing curvature — **PROHIBITED** by the FIREWALL (Principle 2.8.1), or
- (b) An algebraic cross-sector burden coupling not currently present in the framework — would require a new primitive (violating Guardrail 4, "No new primitives without explicit justification").

The **relational burden** channel $\wedge(\text{rel}) = [\hat{C}_\alpha, \Pi_{\text{net}}]$ (§4.4.1) — equivalently, $B_{\text{rel}} = \text{Tr}(\rho \cdot F^\wedge_{\text{cross}})$ (Theorem 1.2.4) — does NOT cross the FIREWALL because it is algebraic (a commutator structure, not a probabilistic average). The relational burden is the derived dark-matter mechanism (§4.4.3); cross-sector gravity via branch weights is quarantined.

§2.9 Architectural Summary

What Section 2 Contains

Subsection	Content	Layer
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§2.0	Purpose: the QM layer	—
§2.1	Reconciliation Principle (variational), $s(a,b)$, $s^*(a,b)$	C
§2.2	Dark energy mechanism (restorative power, Γ_{eff} , ρ_{DE} scaling, 120-order, EDE suppression)	C
§2.3	Fields on A_{red} , burden flux, covariant Laplacian	C

Subsection	Content	Layer
§2.4	Stable modes, particles (low-burden), particle identity	C
§2.5	Mass-burden identity $m^2 \equiv B_0$, gravitational time dilation	C
§2.6	Record sectors, cross-sector burden, stable separation, redundant encoding	C
§2.7	Sectorwise zero-decomposition (M_{red}), Z-envariance, Born rule (STRENGTHENED)	C
§2.8	FIREWALL guardrail, cross-sector gravity quarantine	C

What Section 2 Does NOT Contain (Deferred)

Object	Deferred to	Reason
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K_ω , K_ω^J , Fubini-Study metric	§3	Requires A_{loc} (§3.1, which requires $A_{\text{phy}}^{\text{thin}}$ from §1.7)
Π_{net} (cross-extension network operator)	§3.9	Requires K_ω (§3.1) and sectoral decomposition (§1.2)
Burden tensor $\Theta^{(B)}_{\mu\nu}$	§4.3	Requires the emergent metric (§3)
Three-tier burden decomposition (as gravitational source)	§4.4	Algebraic decomposition is here (§2.3); gravitational sourcing is in §4
Einstein closure, gravity	§4.5	Requires burden tensor (§4.3)
Duration, burden-clock $\alpha(B)$	§4.1	Requires the "now" (§4.2)
Friedmann equation, cosmological application of dark energy	§6	Requires the emergent metric (§3) and the burden tensor (§4.3)

Acyclicity Test

Question: Does Section 2 define every object before using it?

Verification:

- §2.1 uses burden B_Δ (§1.3), events (§1.8), causal order $<$ (§1.8.4). ✓
- §2.2 uses B_Δ , Open Extension (§1.5.3), H (forward-ref to §6.1 — named, not used in definition). ✓
- §2.3 uses M_{red} (§1.4.3), F (§0.3.5), sector projectors (§1.2.3). ✓
- §2.4 uses fields (§2.3.1), B_Δ (§1.3). ✓
- §2.5 uses B_Δ , particle modes (§2.4.2). ✓
- §2.6 uses F_{cross} (§1.2.4), $\hat{C}_\alpha$ (§0.1.6), P_{ij} (§2.6.2). ✓
- §2.7 uses M_{red} (§1.4.3), F (§0.3.5), B_Δ (§1.3), Open Extension (§1.5.3). ✓

- §2.8 uses B_{Δ} , burden linearity (§1.3.3), Θ^{rel} (forward-ref to §4.4.1 — named, not used). ✓

Forward references:

- §2.2.7 references $H(t)$ and the cosmological application (§6) — named as the downstream consequence, not used in the structural scaling derivation.
- §2.3 references X_{ω} (§3) — named as where the field lives, but the field definition uses only A_{red} and M_{red} (available).
- §2.5.2 references §4.1.1 (burden-clock $\alpha(B)$) — named as where the certification lands, not used.
- §2.8.3 references Θ^{rel} (§4.4.1) — named as the relational burden channel, not used in the FIREWALL statement.

No forward reference creates a circularity.

Verdict: Section 2 is acyclic. The architecture holds through the QM layer.

Inline Gap Summary for §2

Gap ID	Location	Description	Severity	Blocks
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GAP-2.1.2	§2.1 Remark 2.1.2	RP derivation from Open Expansion missing (violates Guardrail 4 if irreducible)	FATAL	Primitive count honesty
GAP-2.1.6	§2.1 Theorem 2.1.6	RP well-posedness T-2-conditional	MATERIAL	RP rigor (propagated)
GAP-2.2.7	§2.2 Theorem 2.2.7	Dark energy cosmological predictions conditional on Friedmann (§6.2.2)	MATERIAL	Observational comparison (§6)
GAP-2.3.4	§2.3 Lemma 2.3.4	Anti-hermiticity of burden flux T-2-conditional	MATERIAL	Covariant Laplacian self-adjointness (propagated)
GAP-2.3.7	§2.3 Eq 2.3.7	Wave equation is candidate, not derived	MATERIAL	Field dynamics (T-2/T-3)
GAP-2.7.5	§2.7 Theorem 2.7.5	Born rule full proof (Petz/Bures direction) incomplete	MATERIAL	T-4 closure

Total new gaps surfaced in §2: 6 (1 fatal, 5 material). Of these:

- GAP-2.1.2 is the F3 fix from the patch plan (RP derivation) — the most significant new flag, severity FATAL.
- GAP-2.1.6, GAP-2.3.4 are T-2-propagated dependencies — they resolve when T-2 closes.

- GAP-2.2.7 is the M2 fix (dark energy conditional tags) — resolves when Friedmann is executed (M5).
- GAP-2.3.7 is the wave-equation derivation gap — resolves with T-2/T-3.
- GAP-2.7.5 is the Born rule proof completion — resolves with further development of the Petz/Bures direction.

No NEW structural gaps introduced by the v2.1 rewrite of §2. All 6 gaps are pre-existing from the Phase C program or patch plan, now flagged at their point of origin.

Section 2 is CLOSED. Section 3 (Emergent Geometry) can be merged against this stable QM layer.